

ARJUN PATEL

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PROFESSIONAL SUMMARY

Computational systems biologist with 14 peer-reviewed publications and six years building predictive genome-scale models of microbial metabolism. Specialize in translating multi-omics data into models that guide strain engineering, from *E. coli* proteome allocation to non-model rhizosphere bacteria for bioenergy applications. Built and deployed a web-based platform making ME-model tools accessible to external users. Track record of consulting engagements with academic groups, plus contributions to federally funded research programs (DOE, NSF).

EDUCATION

University of California, San Diego, La Jolla, CA November 2025
Doctor of Philosophy in Bioengineering
GPA: 3.833

Georgia Institute of Technology, Atlanta, GA, May 2019
Bachelor of Science in Biomedical Engineering, minor in Engineering and Business
GPA: 4.0

RESEARCH AND PROFESSIONAL EXPERIENCE

Postdoctoral Scholar, Dr. Cristal Zuniga's Computational Biotechnology Research Group, San Diego State University, January 2026 – Present

- Build ME-models for plant rhizosphere bacteria with integrated pH- and temperature-stress mechanisms using COBRAME, AcidifyME, and FoldME.
- Validate reconstructions against large-scale phenotypic datasets (BIOLOG carbon and nitrogen utilization screens) across pH and temperature gradients.
- Integrate ME-model predictions with multi-omics and biogeochemical data from the Earth Microbiome Project, NMDC, and NEON.
- Write and contribute to federal grant proposals (DOE, NSF) on metabolic modeling of bioenergy-relevant microbial systems.
- Mentor graduate and undergraduate researchers; manage project timelines and deliverables.

Scientific Consultant (Independent Contractor), Self-employed, January 2026 – Present

- Provide computational systems biology consulting to academic research groups under an independent contractor agreement, spanning genome-scale metabolic and ME-model development, multi-omics data analysis, and modeling infrastructure.
- Advise external research groups on constraint-based modeling methods, ME-model reconstruction, and integration of experimental data with computational models.

Graduate Student Researcher, Dr. Bernhard Palsson's Systems Biology Research Group, University of California, San Diego, September 2019 – December 2025

- Worked with machine learning techniques on large-scale multi-omics datasets to elucidate and compare underlying regulatory networks.

- Conducted research in systems biology by integrating multi-omics data, unsupervised machine learning (ICA), and genome-scale metabolism and expression models to study proteome allocation and stress responses in *E. coli*.
- Developed a web-based platform to make modeling and analysis tools accessible to the broader scientific community.
- Collaborated with external research groups to implement constraint-based modeling approaches and integrate experimental data with ME-models, accelerating diverse systems biology projects.

Undergrad Research Assistant, Dr. Denis Tsygankov's Integrative Systems Biology Lab, Georgia Institute of Technology, August 2017 – August 2019

- Developed an algorithm through MATLAB to accurately create a Voronoi graph from a cell cluster image.
- Analyzed the validity of lines and polygons as seeds for Voronoi Graphs in the context of biological cells.
- Created a MATLAB GUI that easily allows a user to upload a cell cluster image and get the resulting Voronoi Graphs.

Undergrad Research Assistant, Biotech/Public Policy with Dr. Aaron Levine, Georgia Institute of Technology, May 2017 – August 2017

- Explored ethical and policy challenges related to the translation of Chimeric Antigen Receptor (CAR) T Cell Therapies.
- Collected and synthesized documents related to the field and major companies involved.
- Co-First Authored a paper on the work completed.

Innovations Intern, Center for Technology Innovation and Commercialization, Wake Forest University, May 2016 – August 2016

- Guided faculty and staff in moving their innovations to the marketplace.
- Structured provisional patent applications for inventions to be protected and licensed to interested companies.
- Engineered and created a novel drug delivery device patented by WIPO (WO2018045287).

PUBLISHED PAPERS

Shin, Jongoh; **Patel, Arjun**; Lou, Xuwen A; Catoi, Edward A; Krishnan, Jayanth; Hefner, Ying; Szubin, Richard; Sung, Jaemin; Son, Hyeoncheol Francis; Zielinski, Daniel C; Palsson, Bernhard O (2026). *A systems-level atlas of carbon-response transcriptional states in Escherichia coli*. Proceedings of the National Academy of Sciences, 123(27), e2531884123.

Patel, Arjun; Banwani, Neha; Mink, Raegan; Prabhakaran, Divya M; Khairnar, Snehal V; Feist, Adam M; Palsson, Bernhard O; Anand, Amitesh (2026). *Aerobicity stimulon in Escherichia coli revealed using multi-scale computational systems biology of respiratory variants*. **iScience**, 29(2), 114715.

Rychel, Kevin; Chen, Ke; Catoi, Edward A; Olson, Connor A; Sandberg, Troy E; Gao, Ye; Xu, Sibe; Hefner, Ying; Szubin, Richard; **Patel, Arjun**; Feist, Adam M; Palsson, Bernhard O (2025).

Laboratory evolution reveals transcriptional mechanisms underlying thermal adaptation of Escherichia coli. **Genome Biol Evol**, 2025 Sep 30;17(10):evaf171.

Kundu, Biki Bapi; Krishnan, Jayanth; Szubin, Richard; **Patel, Arjun**; Palsson, Bernhard O; Zielinski, Daniel C; Ajo-Franklin, Caroline M (2025). *Extracellular respiration is a latent energy metabolism in Escherichia coli.* **Cell**, 188(11), 2907–2924.e23.

Catoiu, Edward A; Krishnan, Jayanth; Li, Gaoyuan; Lou, Xuwen A; Rychel, Kevin; Yuan, Yuan; Bajpe, Heera; **Patel, Arjun**; Choe, Donghui; Shin, Jongoh; Burrows, Joshua; Phaneuf, Patrick V; Zielinski, Daniel C; Palsson, Bernhard O (2025). *iModulonDB 2.0: dynamic tools to facilitate knowledge-mining and user-enabled analyses of curated transcriptomic datasets.* **Nucleic Acids Research**, 53(D1), D99–D106.

Beulig, Felix; Bafna-Rührer, Jonas; Jensen, Peter E; Kim, SH; **Patel, Arjun**; Kandasamy, Vijayalakshmi; Steffen, CS; Decker, Katherine; Zielinski, DC; Yang, Laurence; Ozdemir, Emre; Sudarsan, Suresh; Palsson, Bernhard O (2025). *Trade-off between resistance and persistence in high cell density Escherichia coli cultures.* **mSystems**, 10(7), e0032325.

Patel, Arjun; McGrosso, Dominic; Hefner, Ying; Campeau, Anaamika; Sastry, Anand V; Maurya, Svetlana; Rychel, Kevin; Gonzalez, David J; Palsson, Bernhard O (2024). *Proteome allocation is linked to transcriptional regulation through a modularized transcriptome.* **Nature Communications**, 15(1), 5234.

Dalldorf, Christopher; Rychel, Kevin; Szubin, Richard; Hefner, Ying; **Patel, Arjun**; Zielinski, Daniel C; Palsson, Bernhard O (2024). *The hallmarks of a tradeoff in transcriptomes that balances stress and growth functions.* **mSystems**, 9(7), e00305-24.

Rychel, Kevin; Tan, Justin; **Patel, Arjun**; Lamoureux, Cameron; Hefner, Ying; Szubin, Richard; Johnsen, Josefin; Mohamed, Elsayed Tharwat Tolba; Phaneuf, Patrick V; Anand, Amitesh; Olson, Connor A; Park, Joon Ho; Sastry, Anand V; Yang, Laurence; Feist, Adam M; Palsson, Bernhard O (2023). *Laboratory evolution, transcriptomics, and modeling reveal mechanisms of paraquat tolerance.* **Cell Reports**, 42(9):113105.

Goel, Nikita; Srivastav, Stuti; **Patel, Arjun**; Shirsath, Akshay; Panda, Tushar Ranjan; Patra, Malay; Feist, Adam M; Anand, Amitesh (2023). *TCA cycle tailoring facilitates optimal growth of proton-pumping NADH dehydrogenase-dependent Escherichia coli.* **Microbiology Spectrum**, 11(6), e02225-23.

Anand, Amitesh; **Patel, Arjun**; Chen, Ke; Olson, Connor A; Phaneuf, Patrick V; Lamoureux, Cameron; Hefner, Ying; Szubin, Richard; Feist, Adam M; Palsson, Bernhard O (2022). *Laboratory evolution of synthetic electron transport system variants reveals a larger metabolic respiratory system and its plasticity.* **Nature Communications**, 13(1), 3682.

Anand, Amitesh; Olson, Connor A; Sastry, Anand V; **Patel, Arjun**; Szubin, Richard; Yang, Laurence; Feist, Adam M; Palsson, Bernhard O (2021). *Restoration of fitness lost due to dysregulation of the pyruvate dehydrogenase complex is triggered by ribosomal binding site modifications.* **Cell Reports**, 35(1), 108961.

Zielinski, Daniel Craig; **Patel, Arjun**; Palsson, Bernhard O (2020). *The expanding computational toolbox for engineering microbial phenotypes at the genome scale*. **Microorganisms**, 8(12), 2050.

Imbach, Kathleen; **Patel, Arjun**; Levine, Aaron D (2018). *Ethical considerations in the translation of CAR-T cell therapies*. **Immuno-oncology Insights**.

PUBLISHED PATENTS

Patel, A., 2017. Solid Particulate Measuring Devices, Systems and Methods. WO2018045287

CONFERENCES

Center for Bioenergy Innovation (CBI) 2026 Annual Science Meeting, Asheville, North Carolina

- Delivered an invited flash talk, "*Computational Systems Biology for Rhizosphere Microbiomes*," to the assembled Bioenergy Research Center community and Science & Industry Advisory Board.
- First-authored and presented a poster, "*Genome-Scale Reconstruction and Metabolism-Expression Modeling of Zymomonas mobilis ZM4*," covering ME-model reconstruction (4,627 metabolites, 9,648 reactions, 859 genes), model-predicted proteome allocation validated against LC-MS/MS proteomics, and a host-associated adaptive laboratory evolution (HALE) design for photosynthetic hydrolysate tolerance.

Great Lakes Bioenergy Research Center (GLBRC) 2026 Annual Science Meeting, Lake Geneva, Wisconsin

- First-authored and presented a poster, "*Metabolism and Expression Modeling of Zymomonas mobilis ZM4 and Adaptation to Photosynthetic Hydrolysates*," on ME-model-guided proteome allocation analysis and host-associated adaptive laboratory evolution (HALE) in photosynthetic hydrolysates.

Department of Energy Office of Biological and Environmental Research (BER) 2026 Annual Principal Investigator Meeting, Arlington, Virginia

- Co-authored a poster, "*From Fluxes to Phenotypes: AI-Enhanced Metabolic Modeling of Rhizosphere Bacteria under Changing pH and Temperature*," covering genome-scale and ME-model reconstruction, AI-driven flux analysis, and high-throughput phenotyping of rhizosphere bacteria (*Achromobacter*, *Pseudomonas*, *Variovorax*, *B. subtilis*) under varying pH and temperature.

9th Conference on Constraint-Based Reconstruction and Analysis 2024, San Diego, California

- Hosted and organized a pre-conference workshop on COBRA methods and software developments to facilitate community discussion and accelerate codebase improvements.
- Presented progress and provided a status update on the COBRAME Web Application.

Denmark Technical University Biosustain Annual Seminar 2023, Copenhagen, Denmark

- Presented a poster on the paper, "*Proteome allocation is linked to transcriptional regulation through a modularized transcriptome*. **Nature Communications**, 15(1), 5234."

Denmark Technical University Biosustain Annual Seminar 2021, Copenhagen, Denmark

- Hosted an iModulon workshop to teach the basics of independent component analysis and applications/use cases of iModulons in various organisms.

INVITED TALKS AND GUEST LECTURES

San Diego State University, San Diego, CA

February 24, 2026

- Delivered a guest lecture on the central dogma of molecular biology with a focus on DNA replication for *BIOL 211: Fundamentals of Microbiology* (instructor: Dr. Cristal Zuniga).

Tata Institute of Fundamental Research, Mumbai, India (Remote)

January 6, 2026

- Delivered a follow-up lecture to undergraduate students on systems biology and genome-scale metabolic and expression modeling, expanding on real-world applications and demoing the web-based platform developed to make ME-model tools more accessible to the broader scientific community.

Tata Institute of Fundamental Research, Mumbai, India (Remote)

January 16, 2025

- Delivered a lecture on systems biology and genome-scale metabolic and expression modeling applications to undergraduate students, highlighting approaches for integrating multi-omic data with computational models.

TECHNICAL SKILLS

- **Modeling & Simulation:** Genome-scale metabolic models and reconstructions (GEMs), ME-models (COBRAME, DynamicME, StressME), Flux Balance Analysis (FBA), kinetic/constraint-based modeling.
- **Programming & Data Science:** Python (pandas, NumPy, SciPy, scikit-learn, matplotlib, plotly), MATLAB, Java.
- **Software/Frameworks:** Docker, Git/GitHub, Flask, AWS EC2.
- **Data Analytics:** Multi-omics data integration (transcriptomics, proteomics, metabolomics), statistical modeling, unsupervised machine learning methods (regression, clustering, dimensionality reduction).
- **Other Tools:** SolidWorks, LaTeX, Microsoft Excel/Word/PowerPoint.

TEACHING AND MENTORING

Graduate Teaching Assistant, Bioengineering Department, University of California San Diego, January 2020 – June 2022

- Graded assignments, hosted discussion sections, and managed administrative tasks for Systems Biology and Bioengineering III: Building and Simulating Large-Scale in Silico Models.
- Managed the online forum, graded coursework, and held office hours for ~100 students in Dynamic Simulation in Bioengineering.
- Designed and graded exams, hosted discussion sections, and provided instructional support for Modeling and Computation in Bioengineering class.

Undergraduate Teaching Assistant, Intro to Linear Algebra, Intro to Biomechanics, Georgia Institute of Technology, August 2016 – May 2019

- Led weekly recitations and graded 150+ assignments, quizzes, and exams per week across two courses.

- Earned a 4.97/5.0 teaching evaluation; received the department's Outstanding Student Evaluations TA Award five semesters.

SELECTED AWARDS, HONORS, and SCHOLARSHIPS

Outstanding Student Evaluations Teaching Assistant Award, Math Department at the Georgia Institute of Technology, awarded Fall 2016, Spring 2017, Fall 2017, Spring 2018, Fall 2018

- Awarded to Teaching Assistants who received a 4.9/5.0 or higher with 90% or more feedback on end of year course evaluations.

Alpha Eta Mu Beta Biomedical Engineering Honor Society, Georgia Institute of Technology, awarded Spring 2017

Georgia Tech Honor's Program, Georgia Institute of Technology, awarded Fall 2015

Nell and Spencer Waggnor Scholarship, Winston-Salem Foundation, awarded May 2015

OUTREACH

K–12 Microalgae Outreach, Zuniga Lab, San Diego State University March 6 & 13, 2026

- Co-led a two-part hands-on science module with 5th-grade students at Fletcher Hills Elementary (La Mesa-Spring Valley School District), introducing them to microalgae biology and experimental design.
- Visited the classroom on March 6 to deliver supplies and walk students through a week-long microalgae cultivation experiment.
- Hosted students at the Zuniga Lab on March 13 to analyze their results and learn about laboratory research.

CERTIFICATIONS AND INTERESTS

Certifications

- **CITI Program:** Human Subjects Research / Responsible Conduct of Research.

Interests

- 3rd Degree Black Belt in Tae Kwon Do.
- Proficiency in Clarinet.
- Sports (Basketball, Baseball – avid Charlotte Hornets & San Diego Padres fan)